

# TOWER CRANE DIGITAL TWIN

## Potain MCT 88 — Construction Site Simulation & Capacity Monitoring

Complete study guide: crane anatomy, load-moment physics, capacity\_starter.js,  
STARTER-KIT cross-reference, simulation architecture, constraints, validation,  
digital-twin concept, mentor Q&A bank and glossary

How to use this guide. Chapters 1-2 give you vocabulary. Chapters 3-5 are the physics and code your mentor will probe deepest. Chapters 6-9 map every screen element to the function that produces it. Chapter 11 is a rehearsal bank: cover the answers and recite them. Every number in this guide was taken from the project itself (capacity.js, script.js, CYCLE2\_SUMMARY.md, SOURCES.md, README.md) — nothing is invented.

# Contents

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| Contents . . . . .                                                      | 2         |
| <b>1 Tower cranes in five minutes</b>                                   | <b>4</b>  |
| 1.1 What a tower crane is . . . . .                                     | 4         |
| 1.2 Hammerhead vs flat-top . . . . .                                    | 4         |
| 1.3 Anatomy (every labelled part in the 3D scene) . . . . .             | 4         |
| 1.4 The three motions (verbs your mentor expects) . . . . .             | 5         |
| <b>2 The real crane: Potain MCT 88</b>                                  | <b>5</b>  |
| 2.1 Identity . . . . .                                                  | 5         |
| 2.2 Verified specification (published data only) . . . . .              | 6         |
| 2.3 Source trail (answer 'where is this from?') . . . . .               | 6         |
| <b>3 Load-moment physics (the heart of the project)</b>                 | <b>6</b>  |
| 3.1 Overturning moment . . . . .                                        | 6         |
| 3.2 The two-piece model (STARTER-KIT section 1) . . . . .               | 6         |
| 3.3 Utilisation and risk thresholds . . . . .                           | 7         |
| <b>4 capacity_starter.js, line by line</b>                              | <b>7</b>  |
| 4.1 The MCT88 data object . . . . .                                     | 8         |
| 4.2 tipLoadAt() — linear interpolation between table rows . . . . .     | 8         |
| 4.3 capacityAt() — the implemented TODO . . . . .                       | 8         |
| 4.4 validate() — the harness . . . . .                                  | 8         |
| 4.5 hookInZone() — point-in-rectangle . . . . .                         | 8         |
| <b>5 STARTER-KIT.md cross-reference</b>                                 | <b>8</b>  |
| <b>6 How the simulation runs</b>                                        | <b>9</b>  |
| 6.1 Files and boot . . . . .                                            | 9         |
| 6.2 State, sliders and the frame loop . . . . .                         | 9         |
| 6.3 hookWorld() — where the load is . . . . .                           | 10        |
| 6.4 What is real, calculated, simulated (memorise this table) . . . . . | 10        |
| <b>7 Risk, zones and clearances</b>                                     | <b>10</b> |
| 7.1 The risk badge pipeline . . . . .                                   | 10        |
| 7.2 Exclusion zones (geometry, not physics) . . . . .                   | 11        |
| 7.3 Building clearance: two levels . . . . .                            | 11        |
| <b>8 The automatic lift (10 stages)</b>                                 | <b>12</b> |
| <b>9 Validation deep-dive</b>                                           | <b>12</b> |
| 9.1 The three points and the console evidence . . . . .                 | 12        |
| 9.2 Hand derivation (write this on paper too) . . . . .                 | 12        |
| 9.3 Known discrepancy (shows honesty, expect this question) . . . . .   | 12        |
| <b>10 Digital twin: concept and honest scope</b>                        | <b>13</b> |

|           |                                            |           |
|-----------|--------------------------------------------|-----------|
| <b>11</b> | <b>Mentor Q&amp;A; bank</b>                | <b>13</b> |
| <b>12</b> | <b>Glossary, sources, demo checklist</b>   | <b>15</b> |
| 12.1      | Glossary (30-second definitions) . . . . . | 15        |
| 12.2      | Sources . . . . .                          | 15        |
| 12.3      | Three-minute demo checklist . . . . .      | 16        |

Conventions: t = tonnes, m = metres, t·m = tonne-metres (moment). r = working radius. Code listings are ASCII simplifications; the real files use the same logic.

# 1 Tower cranes in five minutes

## 1.1 What a tower crane is

A tower crane is a fixed, tall lifting machine used to raise and place heavy materials (steel, concrete panels, machinery) across a construction site. Unlike a mobile crane, it is erected on site, anchored to a foundation, and stays for months. Its long horizontal arm lets one machine cover a whole site by combining three motions: trolley travel (in/out along the arm), slewing (rotating the whole top), and hoisting (raising/lowering the hook).

## 1.2 Hammerhead vs flat-top

Classic hammerhead cranes carry an A-frame tower above the arm with long tie-bars. A flat-top crane — like the Potain MCT 88 — has no tower head: the jib and counter-jib are cantilevered directly from the slewing unit with short pendant bars to a low apex. Flat-tops are faster to erect, need less height above the jib (useful beside buildings and under other cranes), and are the modern standard for building construction.

## 1.3 Anatomy (every labelled part in the 3D scene)

### Potain MCT 88 (flat-top): side view, main parts (not to scale)

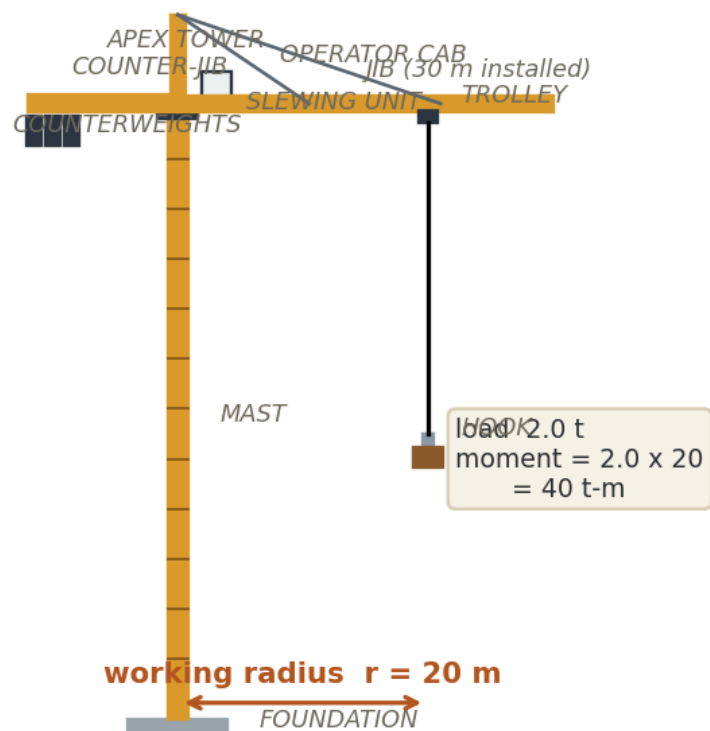


Figure 1 — Flat-top tower crane, side view. Arrow shows the working radius; the box shows load x radius = overturning moment.

| Part              | What it is                                           | In the simulation                                                   |
|-------------------|------------------------------------------------------|---------------------------------------------------------------------|
| Foundation / base | Concrete pad + fixing angles anchoring the crane     | Grey pad at origin, inside the base exclusion zone                  |
| Mast              | Lattice tower of stacked sections; gives hook height | 12 yellow lattice sections (~48 m); TALL-MAST config is visual only |
| Slewing unit      | Ring + motors rotating the entire top of the crane   | Rotates with the SLEW slider (-180 deg to +180 deg)                 |

| Part                         | What it is                                                    | In the simulation                                                   |
|------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|
| Operator cab                 | Control cabin with the operator                               | White cabin beside the slew unit (decorative)                       |
| Apex tower                   | Short tower carrying the pendant bars (flat-tops keep it low) | Yellow box above the slew unit                                      |
| Jib                          | Horizontal working arm the trolley runs along                 | 30 m installed (slider 20-53 m rebuilds it; envelope ring rescales) |
| Counter-jib + counterweights | Short rear arm + concrete blocks balancing the jib load       | Modelled; labels shown in scene                                     |
| Trolley                      | Motorised carriage travelling along the jib; sets the radius  | TROLLEY slider, clamped 4 m to (jib - 1) m                          |
| Hook + load                  | Hook block on wire rope; the carried material                 | HOIST slider sets hook height; sway is simulated wind effect        |
| Pendant bars                 | Steel ties from apex to jib/counter-jib                       | Rebuilt when jib length changes                                     |

## 1.4 The three motions (verbs your mentor expects)

- Hoist / lower — wire rope winds on the hoist drum; hook height changes. Slider 2-46 m; clamped below the mast top.
- Trolley in / out — carriage runs along the jib; this sets the working radius  $r$ , which is why capacity changes when you move it.
- Slew left / right — whole top rotates about the mast; moves the hook around the site without changing radius or height.

Key sentence to memorise: radius sets the overturning moment (load  $\times$  radius), so trolleying OUT at constant load always reduces the safety margin. Slewing and hoisting move the load but do not change the load-moment utilisation by themselves.

# 2 The real crane: Potain MCT 88

## 2.1 Identity

The simulated crane is the Potain MCT 88 (EN 14439 C25), a top-slewing flat-top tower crane built by Potain / The Manitowoc Company, Inc. EN 14439 is the European crane-design standard; C25 is the crane's classification group within it. FEM 1.001 is the European materials-handling federation rule set the data sheet follows.

## 2.2 Verified specification (published data only)

| Item                            | Value                                                                                                   | Origin                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Maximum rated capacity          | 5 t (flat inner region)                                                                                 | MCT 88 sheet, upper 5.5-USt line (= 4.99 t, used as 5 t model) |
| Installed jib (validation)      | 30 m                                                                                                    | Chosen configuration for Cycle 2                               |
| Maximum jib of the model        | 52 m max radius (170.6 ft)                                                                              | Model upper limit - NOT the installed config                   |
| Tip load at 30 m                | 2.70 t                                                                                                  | Published tip-load table                                       |
| Load moment M                   | 81 t·m = 2.70 x 30                                                                                      | Calculated, Chapter 3                                          |
| Corner radius                   | 16.2 m = 81 / 5                                                                                         | Calculated, Chapter 3                                          |
| Tip-load table (jib m to tip t) | 20 to 3.95 - 25 to 3.00 - 30 to 2.70 - 35 to 2.27 - 40 to 1.95 - 45 to 1.70 - 50 to 1.45 - 53.1 to 1.15 | STARTER-KIT section 2 (real brochure figures)                  |

## 2.3 Source trail (answer 'where is this from?')

- Interactive catalogue: Potain e-catalogue EN 14439 C25-D25 (official Manitowoc site). It blocks bots/scripts, so it was verified in a normal browser session (SOURCES.md).
- Load-chart document: Potain MCT 88 Data Sheet, FEM 1.001-A3 (Code 09-003-.5M-0610), 'Load charts' page, 98 ft jib row (about 29.87 m, taken as the 30 m nominal), UPPER line (2-fall, max 5.5 USt). Read via the cranenetwork.com mirror of the same Manitowoc sheet.
- Raw readings (imperial, upper 98 ft row): 66 ft to 4.3 USt, 82 ft to 3.3 USt, 98 ft to 2.8 USt. Conversions: 1 ft = 0.3048 m, 1 USt = 0.907185 t, giving 20.1 m to 3.90 t, 25.0 m to 2.99 t, 29.9 m to 2.54 t.
- Rule: never cite an AI response as a source. All chartT values were read from the manufacturer sheet.

30 m vs 52 m (classic confusion): 30 m is the installed jib chosen for validation; 52 m is the crane model's maximum jib. The slider (20-53 m) explores other installed lengths by interpolating the tip table.

# 3 Load-moment physics (the heart of the project)

## 3.1 Overturning moment

A load hung at radius  $r$  tries to tip the crane over with moment  $M = \text{load} \times r$  (tonne-metres). The crane resists with its counterweights and base. Safe operation means keeping the applied moment within the rated envelope — that envelope is the load chart.

## 3.2 The two-piece model (STARTER-KIT section 1)

The starter kit idealises the chart as two pieces. First compute the crane's moment constant from the published tip:  $M = \text{tipLoad} \times \text{jibLength} = 2.70 \times 30 = 81 \text{ t}\cdot\text{m}$ . Then the corner radius where the flat part meets the curve:  $\text{corner} = M / \text{maxCapacity} = 81 / 5 = 16.2 \text{ m}$ .

| Radius $r$                  | capacityAt(MCT88, 30, $r$ ) | Why                                   |
|-----------------------------|-----------------------------|---------------------------------------|
| $r \leq 0$                  | 5 t                         | Degenerate input returns max capacity |
| $0 < r \leq 16.2 \text{ m}$ | 5 t flat                    | Structure-limited inner region        |

| Radius r                     | capacityAt(MCT88, 30, r) | Why                                       |
|------------------------------|--------------------------|-------------------------------------------|
| $16.2 < r \leq 30 \text{ m}$ | $81 / r$                 | Moment-limited: constant 81 t·m hyperbola |
| $r = 30 \text{ m (tip)}$     | 2.70 t                   | Curve MUST pass through the published tip |
| $r > 30 \text{ m (jib)}$     | null (out of range)      | Beyond installed jib: no chart value      |

Check three points by hand:  $r = 10$  gives 5 t (flat).  $r = 20$  gives  $81/20 = 4.05 \text{ t}$ .  $r = 25$  gives  $81/25 = 3.24 \text{ t}$ . The model never returns more than maxCapacity.

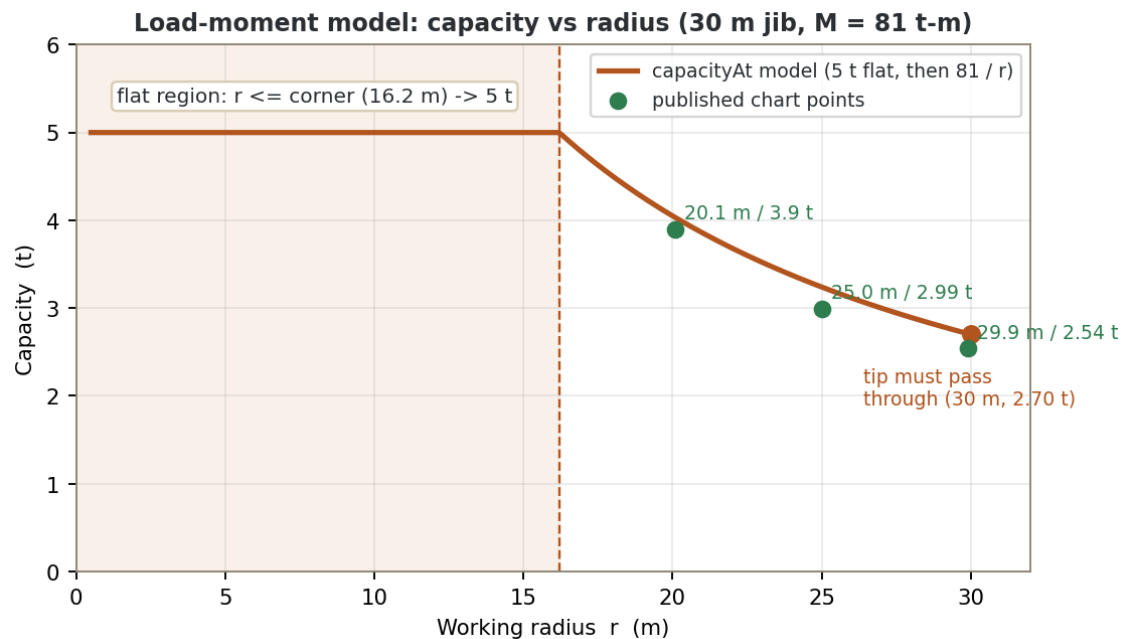


Figure 2 — The two-piece curve: flat 5 t to the 16.2 m corner, then the  $81/r$  hyperbola through the tip. Green dots are the published check points.

### 3.3 Utilisation and risk thresholds

Utilisation  $u = \text{load} / \text{capacity}(r)$ . It is the single number the badge shows. Thresholds (a starter-kit wiring choice, NOT a manufacturer limit):  $u < 0.90$  SAFE,  $0.90 \leq u < 1.00$  WARNING,  $u \geq 1.00$  (or out of range) DANGER. Example: 2.0 t at 20 m: capacity 4.05 t,  $u = 49.4\%$  SAFE. Same load at 29 m: capacity 2.79 t,  $u = 71.6\%$  SAFE. A 4.5 t load at 20 m:  $u = 111\%$  DANGER.

Capacity constraint vs geometric constraint: capacity (load-moment) is physics — can the crane hold it? Exclusion zones and building clearance are geometry — may the load be there? They are computed independently and shown separately. Never confuse them.

## 4 capacity\_starter.js, line by line

The mentor's scaffold file (capacity.starter.js) provided the MCT88 tip table and left one TODO — implement capacityAt per section 1. The project ships it as capacity.js with identical names and logic, exposed as globals (var + window) instead of ES-module export so index.html keeps working when opened directly via file://, where modules are blocked.

## 4.1 The MCT88 data object

```
var MCT88 = {
  label: 'Potain MCT 88 (C25)',
  maxCapacityT: 5,
  tipLoadByJib: [           // installed jib (m) -> published tip (t)
    { jibM: 20, tipT: 3.95 }, { jibM: 25, tipT: 3.0 },
    { jibM: 30, tipT: 2.7 }, { jibM: 35, tipT: 2.27 },
    { jibM: 40, tipT: 1.95 }, { jibM: 45, tipT: 1.7 },
    { jibM: 50, tipT: 1.45 }, { jibM: 53.1, tipT: 1.15 } ] };
// Every row is a real brochure figure (STARTER-KIT section 2).
```

## 4.2 tipLoadAt() — linear interpolation between table rows

For a jib length between two rows, the tip load is interpolated: fraction  $f = (jib - a.jib) / (b.jib - a.jib)$ ,  $tip = a.tip + f * (b.tip - a.tip)$ . Clamped at both ends. Worked example (do this on paper): jib 27.5 m sits between 25 (3.00 t) and 30 (2.70 t):  $f = 2.5 / 5 = 0.5$ ,  $tip = 3.00 + 0.5 * (2.70 - 3.00) = 2.85$  t. Then  $M = 2.85 \times 27.5 = 78.375$  t·m, corner =  $78.375 / 5 = 15.675$  m. This is exactly how the jib slider re-derives the model.

## 4.3 capacityAt() — the implemented TODO

```
function capacityAt(model, jibM, radiusM) {
  if (radiusM <= 0) return model.maxCapacityT; // degenerate -> max
  if (radiusM > jibM) return null;           // beyond jib -> no chart
  var M = tipLoadAt(model, jibM) * jibM;     // moment constant
  var cornerR = M / model.maxCapacityT;      // flat/hyperbola joint
  if (radiusM <= cornerR) return model.maxCapacityT;
  return M / radiusM;                        // hyperbola, never exceeds max
}
```

## 4.4 validate() — the harness

`validate(model, jibM, chart, tolerancePct)` loops over published (radiusM, chartT) points, computes the model value, prints  $errorPct = |model - chart| / chart \times 100$  per point, and returns the worst error with PASS if worst  $\leq$  tolerance (10). The console output at boot is part of the evidence (Chapter 9 reproduces it).

## 4.5 hookInZone() — point-in-rectangle

```
function hookInZone(hookX, hookZ, zone) { // zone = {x0,z0,x1,z1}
  return hookX >= Math.min(zone.x0, zone.x1) && hookX <= Math.max(zone.x0, zone.x1)
    && hookZ >= Math.min(zone.z0, zone.z1) && hookZ <= Math.max(zone.z0, zone.z1);
}
// Edges count as inside. Circle zones (Crane-2) use distance < r instead.
```

Truth table (base rect x[-9,9] z[-9,9]): (0,0) inside TRUE; (20,20) outside FALSE; (9,-9) on the edge TRUE. The check runs every frame on the hook ground point from `hookWorld()`.

# 5 STARTER-KIT.md cross-reference

The starter kit is the mentor's specification. The original file is not in this repository — verify the section numbers below against your copy. Everything the project claims from it is traceable:

| Section (as cited) | Content                                    | Where it is used                                                      |
|--------------------|--------------------------------------------|-----------------------------------------------------------------------|
| Section 1          | Two-piece load-moment rule + corner radius | <code>capacityAt()</code> (Ch.4.3); physics card in Digital Twin view |
| Section 2          | Tip-load table, real brochure figures      | <code>MCT88.tipLoadByJib</code> ; spec table (Ch.2.2)                 |
| Section 4          | Official Potain e-catalogue link           | Crane Data source table; SOURCES.md                                   |

| Section (as cited) | Content                                                  | Where it is used                                    |
|--------------------|----------------------------------------------------------|-----------------------------------------------------|
| Section 6          | Exclusion-zone check via hookInZone on hook ground point | zoneHit() every frame; red highlight; banner (Ch.7) |

If your mentor's copy differs, the code comments in capacity.js and the SOURCES.md trail are the authoritative record of what was actually implemented.

## 6 How the simulation runs

### 6.1 Files and boot

- index.html — layout shell; loads Three.js r128 (non-module CDN builds, file:// safe), then capacity.js, script.js, ui-mirror.js.
- capacity.js — real data + model (Chapter 4). No DOM, no rendering.
- script.js — all of the 3D scene, state, checks and wiring (single IIFE).
- ui-mirror.js — read-only display mirrors (sidebar forwarding, status pills). Zero physics; fully guarded.
- Boot order: createSite, createBuildings, createCrane, wireUI, refreshFloors, camera preset, buildValidation, drawChart, validate() to console, then the animateCrane() loop.

### 6.2 State, sliders and the frame loop

Live state lives in S: hoist/trolley/slew actual vs targets (tHoist/tTrolley/tSlew), load, jibLen, wind, modes, cap/util/moment/corner, exclHit, bldgHit/bldgNear, risk. Sliders write targets; each frame updateCraneMovement() eases actual values toward them (speed-scaled), updateCollisionDetection() recomputes capacity + zones, updateUI() repaints text. Trolley clamps to 4 m ... (jib-1); hoist clamps below the mast top (48 m, 56 m tall-mast).

## 6.3 hookWorld() — where the load is

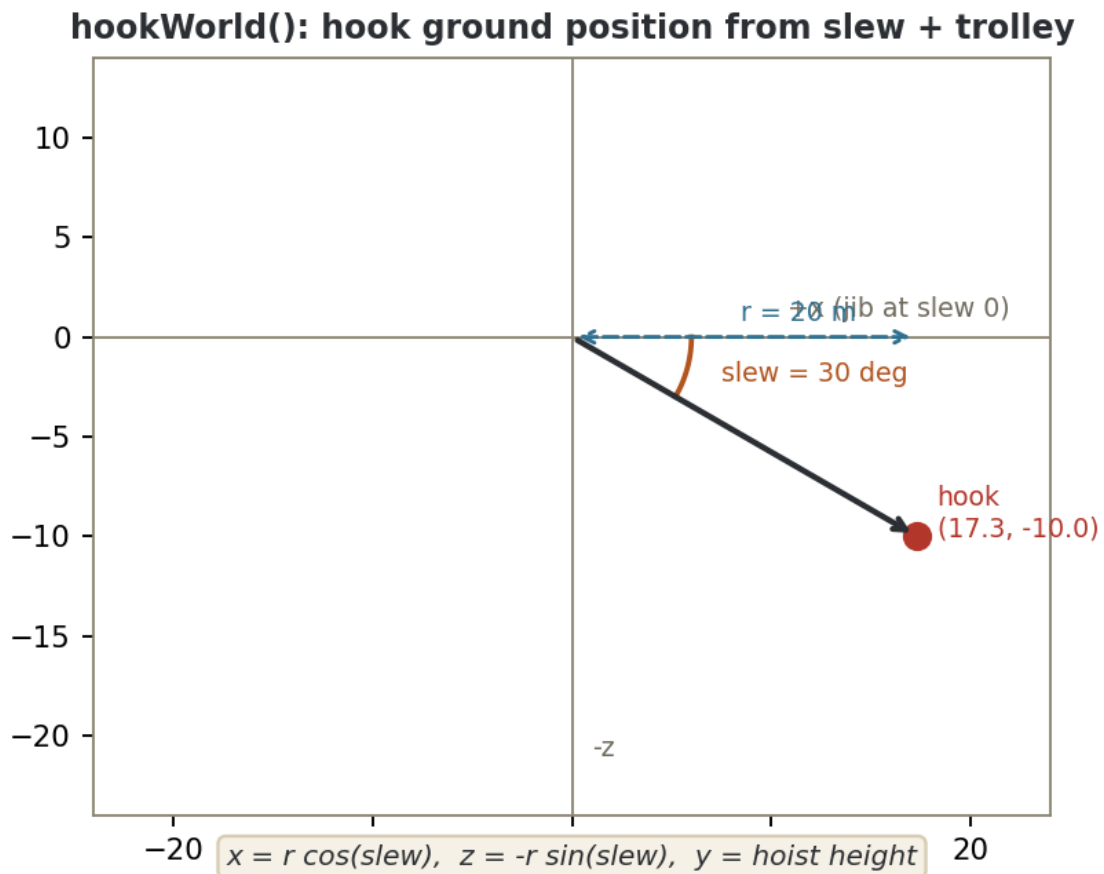


Figure 3 — Hook ground position from slew angle and trolley radius. Example: slew 30 deg, r 20 m gives (17.3, -10.0).

With slew angle  $a$  (degrees) and trolley radius  $r$ :  $x = r \cdot \cos(a)$ ,  $y = \text{hoist height}$ ,  $z = -r \cdot \sin(a)$ . The code comments note the three.js rotation mapping is approximated for display consistency — say this if asked, it shows you read the source. The load centre tested against zones/buildings is the hook point shifted  $\sim 1$  m down.

## 6.4 What is real, calculated, simulated (memorise this table)

| Class                     | Items                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------|
| REAL (manufacturer / kit) | MCT 88 identity, 5 t max, tip table, chart points, zone-test math, EN 14439/FEM refs               |
| CALCULATED                | M, corner, capacity(r), utilisation, validation errors, zone in/out                                |
| SIMULATED                 | 3D crane/buildings/site, hoist/trolley/slew animation, wind sway + arrows, lift sequencing, speeds |
| NOT IMPLEMENTED           | Sensors, physical link, structural/dynamics analysis, certified lift planning                      |

# 7 Risk, zones and clearances

## 7.1 The risk badge pipeline

Every frame:  $\text{cap} = \text{capacityAt}(\text{MCT88}, \text{jibLen}, \text{trolley})$ ;  $\text{util} = \text{load}/\text{cap}$  (999 if out of range, forcing DANGER). Base risk from util thresholds; then building proximity upgrades it: inside-volume conflict forces high, early-warning approach upgrades low to mid. The banner shows the first conflict plus a count ('+2 more').

## 7.2 Exclusion zones (geometry, not physics)

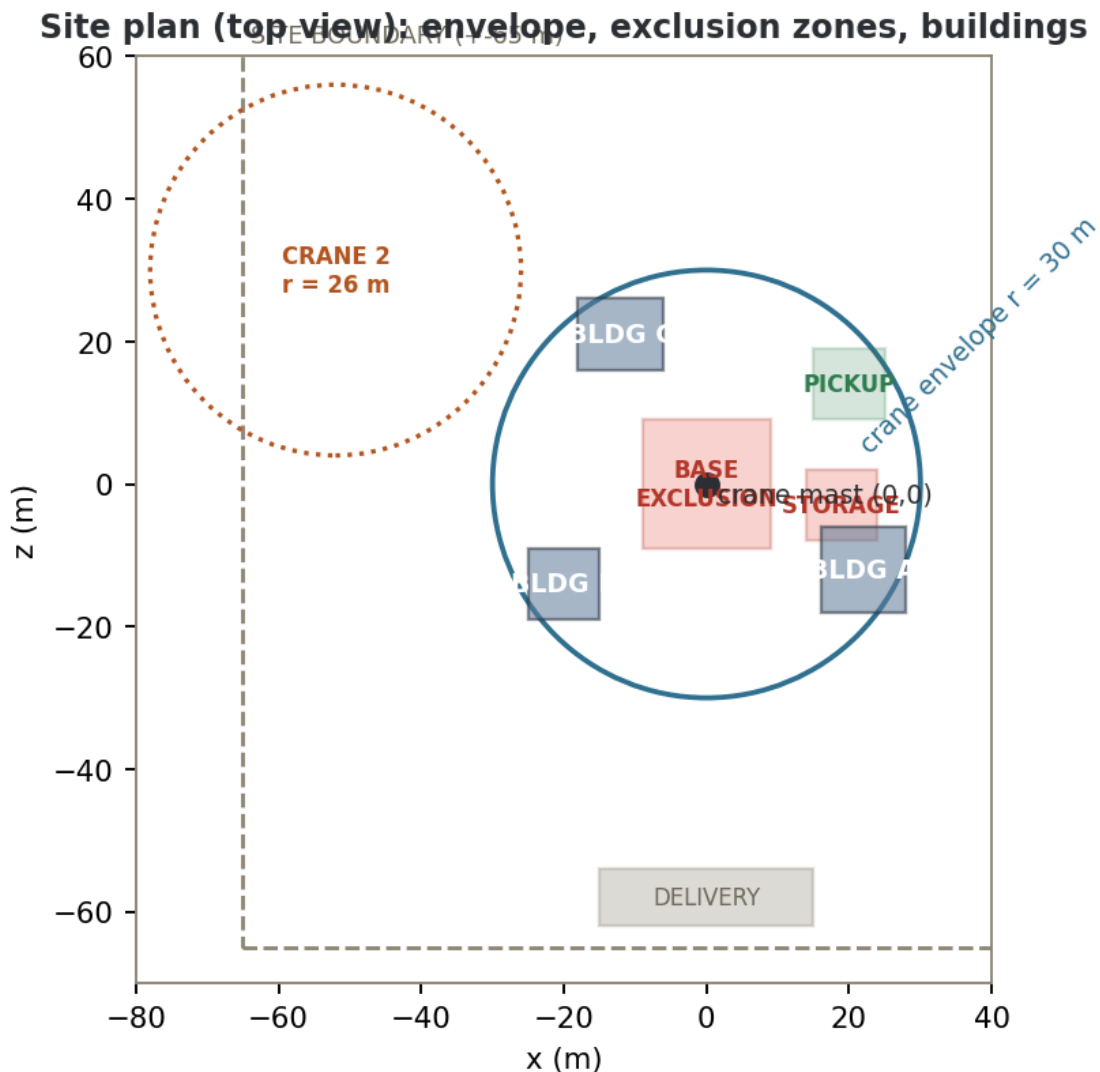


Figure 4 — Site plan: 30 m envelope, base + storage exclusion rects, pickup pad, delivery strip, Crane-2 circle, buildings A/B/C. Fence at +65 m.

| Zone                   | Geometry                                         | Result when hook/load inside             |
|------------------------|--------------------------------------------------|------------------------------------------|
| Crane base             | Rect $x[-9,9]$ $z[-9,9]$ , only if hook $y < 12$ | EXCLUSION ZONE VIOLATION + red highlight |
| Storage                | Rect $x[14,24]$ $z[-8,2]$                        | VIOLATION + red highlight                |
| Crane-2 interference   | Circle $(-52,30)$ $r = 26$ m (documented extra)  | VIOLATION (amber disc turns red)         |
| Pickup / delivery pads | Visual only                                      | NOT enforced (say this!)                 |

## 7.3 Building clearance: two levels

Simplified box test per building with contact tolerance 0.8 m: (1) **CONFLICT** when inside the volume (red, DANGER, auto-lift holds); (2) **APPROACH WARNING** inside the 4 m horizontal / +5 m above-roof buffer (amber, fires BEFORE contact; auto-lift holds during travel but allows the final LOWER/PLACE approach). Buffers are example values — label them simulated, never certified. In code these are the constants `BLDG_WARN_XY = 4.0`, `BLDG_WARN_TOP = 5.0` and the contact tolerance `BLDG_INSIDE_E = 0.8`, consumed by `buildingClearance()` into `S.bldgNear` / `S.bldgHit`.

Demo line: 'Capacity answers CAN it hold it; zones answer MAY it be there. Two independent checks, two separate screen sections.'

## 8 The automatic lift (10 stages)

| Step            | What happens                         | Gate before advancing     |
|-----------------|--------------------------------------|---------------------------|
| 1 PICKUP        | Travel to the green pad polar target | Position reached          |
| 2 HOIST (lower) | Hook down to 2.5 m                   | Position reached          |
| 3 RIGGING       | Load attaches (simulated)            | Timer                     |
| 4 HOIST UP      | Lift to safe travel height           | Capacity + zone checks    |
| 5 TROLLEY OUT   | Radius to target                     | Capacity + zone checks    |
| 6 SLEW          | Rotate toward target building        | Capacity + zone checks    |
| 7 CHECKS        | Capacity + exclusion verified        | Full autoChecks()         |
| 8 LOWER         | Descend to floor target + 1.2 m      | Checks (approach allowed) |
| 9 PLACE         | Load detaches (simulated)            | Timer                     |
| 10 RETURN       | Hook back to travel height           | Complete -> manual mode   |

Travel height =  $\min(\text{mastTop} - 2, \max(\text{target} + 10, 34))$  — high enough to clear roofs (simulated rule). Any slider input during auto mode drops back to manual. autoChecks() holds on DANGER, zone violation, building conflict, or early-warning approach (except LOWER/PLACE). The on-screen live text each stage (LOAD/R/CAP/UTIL/zone/building) is the lift's flight recorder — point at it in the demo.

## 9 Validation deep-dive

### 9.1 The three points and the console evidence

| Radius | Published (chartT) | Model  | Error | Status |
|--------|--------------------|--------|-------|--------|
| 20.1 m | 3.90 t             | 4.03 t | 3.3%  | PASS   |
| 25.0 m | 2.99 t             | 3.24 t | 8.4%  | PASS   |
| 29.9 m | 2.54 t             | 2.71 t | 6.7%  | PASS   |

Worst error  $8.4\% \leq 10\% \rightarrow \text{PASS}$ . Reproduce any time in DevTools: `validate(MCT88, 30, CHART30)`. The table on screen is built by `buildValidation()` from the same `capacityAt()` that drives the sim — model and display cannot disagree.

### 9.2 Hand derivation (write this on paper too)

30 m jib: tip 2.70 t  $\rightarrow M = 2.70 \times 30 = 81 \text{ t}\cdot\text{m} \rightarrow \text{corner} = 81/5 = 16.2 \text{ m} \rightarrow \text{capacity}(20 \text{ m}) = 81/20 = 4.05 \text{ t}$ . Same text ships in `validation-calculation.txt`; the photo of your handwriting is deliverable evidence, not a replacement for the chart.

### 9.3 Known discrepancy (shows honesty, expect this question)

The sheet's own 98 ft tip reads 2.8 USt = 2.54 t, while the metric tip table says 2.70 t at 30 m — a ~6% gap from different standard editions/rounding (ASME vs FEM). It is reported in `SOURCES.md` and `CYCLE2_SUMMARY $P`, not hidden. Likewise the  $\pm 10\%$  criterion passes partly because the model is an idealisation; say so.

## 10 Digital twin: concept and honest scope

What this prototype connects (and what is still simulated)

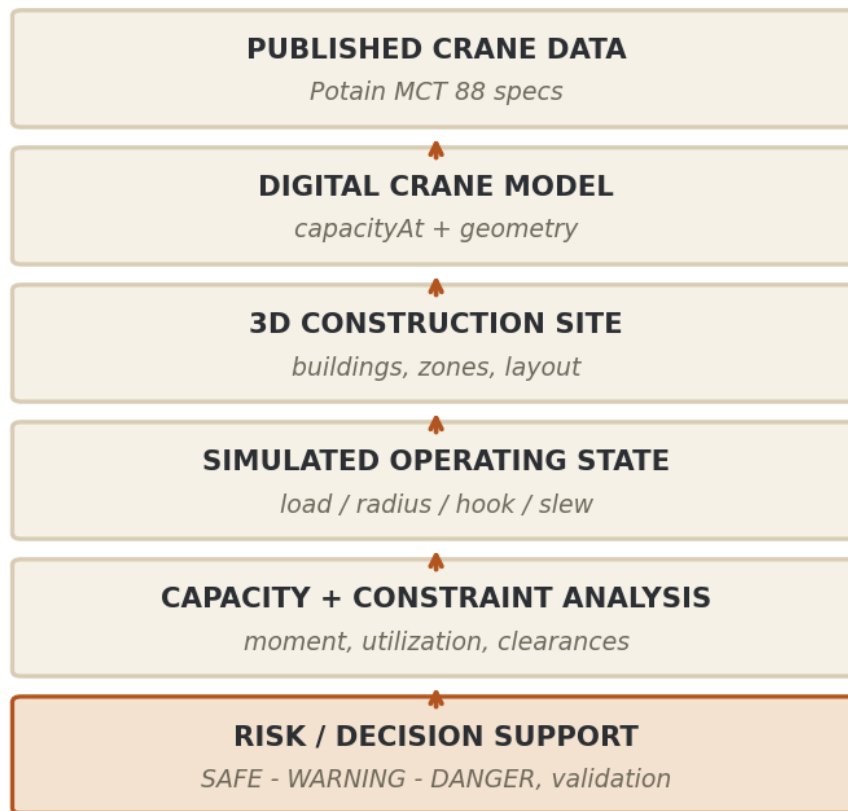


Figure 5 — What this prototype connects. The bottom box is the honest status: a data-driven simulation/prototype, not a live twin.

Definitions to recite: BIM = building/site information (geometry + context). Simulation = behaviour model (movement + capacity). Digital twin = representation + live state + analysis kept in sync with the physical asset. This project demonstrates the first two and the analysis; the live-sync loop (sensors, IoT feed, feedback to operations) is explicitly missing — hence 'prototype using simulated operating inputs, not connected to a live physical crane or IoT sensor feed.'

## 11 Mentor Q&A bank

Twenty-four likely questions. Cover the right column and rehearse aloud.

### Q1. Why a flat-top (MCT 88) instead of a hammerhead?

No tower head above the jib: faster erection, less overhead height near buildings and other cranes. Standard for modern building sites.

### Q2. What do 5 t, 2.70 t, 81 t·m, 16.2 m each mean?

Max rated capacity; published tip load at the 30 m tip; moment constant  $M = 2.70 \times 30$ ; corner radius 81/5 where the flat region meets the hyperbola.

### Q3. Derive the corner radius from scratch.

M comes from the published tip ( $2.70 \times 30 = 81$ ). The flat region ends where the hyperbola would exceed max capacity:  $M/r = 5$  gives  $r = 81/5 = 16.2$  m.

#### Q4. Why is capacity flat, then hyperbolic?

Inner region is structure-limited (5 t constant); outer region is moment-limited (constant 81 t-m, so capacity = 81/r).

#### Q5. What does capacityAt return beyond the jib, and why null not zero?

null = 'no chart value / out of range'. Zero would read as a valid (overturned) capacity and corrupt utilisation; null forces the DANGER path instead.

#### Q6. Compute tipLoadAt for a 27.5 m jib.

Between 25 (3.00) and 30 (2.70):  $f = 0.5$ , tip = 2.85 t,  $M = 78.375$  t-m, corner = 15.675 m.

#### Q7. How is validation error computed? What is the result?

$|model - chart| / chart \times 100$  per point: 3.3%, 8.4%, 6.7%. Worst 8.4%  $\leq 10\%$  so PASS. Same capacityAt drives the sim.

#### Q8. Why is there any error at all?

The two-piece curve is the starter kit's teaching idealisation, not the manufacturer's equation; plus imperial-to-metric conversion noise.

#### Q9. Explain the 2.54 t vs 2.70 t tip discrepancy.

Sheet's 98 ft tip (2.8 USt = 2.54 t) vs metric tip-table 30 m (2.70 t): ~6% gap from different editions/rounding (ASME vs FEM). Reported, not hidden.

#### Q10. What is hookInZone and what feeds it?

Point-in-rectangle test on the hook ground point from hookWorld(), run every frame; edges count as inside.

#### Q11. Capacity vs exclusion zone — aren't both 'safety'?

Different physics: capacity = load-moment (CAN it hold it); zones = geometry (MAY it be there). Independent checks, separate UI.

#### Q12. What are the risk thresholds and whose are they?

$<0.90$  SAFE,  $0.90-<1.00$  WARNING,  $\geq 1.00$  or out-of-range DANGER. A starter-kit wiring choice, not a manufacturer limit.

#### Q13. How does building proximity change risk?

Inside volume forces DANGER; 4 m / +5 m buffer approach upgrades SAFE to WARNING. Buffers are simulated example values.

#### Q14. What happens if the hook enters the storage rect?

EXCLUSION ZONE VIOLATION: banner, red highlight, constraints text; auto-lift holds until slew/radius clears it.

#### Q15. Why does auto-lift hold on approach but allow LOWER near a building?

Travel stages must never touch; the final LOWER/PLACE is the intentional delivery, so the hold is lifted there by design.

#### Q16. Write hookWorld for slew 30 deg, trolley 20 m, hoist 24 m.

$x = 20\cos 30 = 17.3$ ,  $y = 24$ ,  $z = -20\sin 30 = -10.0$ . Mapping is display-approximated per code comment.

#### Q17. Why does moving the trolley change the badge but slewing doesn't?

Trolley changes  $r$  hence capacity and utilisation; slew/hoist move position without changing the moment ratio (except zone/building effects).

### Q18. What is simulated vs real here?

Real: identity, 5 t, tips, chart points, zone-test math. Calculated: M, corner, capacity, utilisation, errors. Simulated: 3D scene, motion, wind, sequencing. Missing: sensors, live sync, certified planning.

### Q19. Why is wind only an indicator?

Sway amplitude (0.15/0.5/1.4) and arrows visualise LOW/MODERATE/HIGH; there is no aerodynamic or certified wind analysis.

### Q20. Why globals instead of ES modules in capacity.js?

Identical logic/names to capacity.starter.js, but file:// blocks modules — globals keep double-click-to-open working.

### Q21. Why Three.js r128 non-module builds?

Classic scripts run without a server or import map; OrbitControls attaches to the same global THREE.

### Q22. What does the 30 m envelope ring show?

Operating radius = installed jib; it rescales with the jib slider. Pickup/delivery visuals are not enforced zones.

### Q23. What is still missing for a real digital twin?

Continuous physical-to-virtual sync, real sensor/ops data, validated dynamics, live site state, feedback/decision loop.

### Q24. Where is each number sourced?

SOURCES.md: e-catalogue + FEM 1.001-A3 sheet via mirror, conversions shown (1 ft = 0.3048 m, 1 USt = 0.907185 t), accessed 2026-09-11.

## 12 Glossary, sources, demo checklist

### 12.1 Glossary (30-second definitions)

| Term                 | Meaning                                                              |
|----------------------|----------------------------------------------------------------------|
| Jib                  | Horizontal working arm; trolley radius runs along it                 |
| Trolley              | Carriage on the jib; its position IS the working radius              |
| Slew                 | Rotation of the crane top about the mast                             |
| Hoist                | Rope winding that sets hook height                                   |
| Load moment          | load x radius (t-m); the overturning demand                          |
| Corner radius        | Joint of flat and hyperbolic capacity regions (16.2 m)               |
| Utilisation          | load / allowed capacity; drives the badge                            |
| Exclusion zone       | Forbidden ground region tested by hookInZone                         |
| Building clearance   | Simplified box test; conflict vs approach warning                    |
| BIM                  | Building/site information model (geometry + context)                 |
| Digital twin         | Synced virtual + physical pair (this project: prototype of one half) |
| FEM 1.001 / EN 14439 | European crane design rules / crane standard (C25 group)             |

### 12.2 Sources

- Potain e-catalogue EN 14439 C25-D25 (official Manitowoc) + MCT 88 Data Sheet FEM 1.001-A3 via cranenetwork mirror — full trail in SOURCES.md.

- Project files: capacity.js, script.js, index.html, style.css, ui-mirror.js, README.md, CYCLE2\_SUMMARY.md, validation-calculation.txt.
- STARTER-KIT.md / capacity.starter.js: mentor originals (not in repo) — sections 1, 2, 4, 6 as cross-referenced in Chapter 5.

### 12.3 Three-minute demo checklist

- 0:00 — Sidebar brand + 3D scene: name the parts (Ch.1).
- 0:30 — Load slider to DANGER: read utilisation math aloud (Ch.3).
- 1:00 — Trolley across the 16.2 m corner: flat then falling capacity (Fig.2).
- 1:30 — Slew into storage: violation banner + red zone (Fig.4).
- 2:00 — START DEMONSTRATION LIFT: narrate stages + holds (Ch.8).
- 2:30 — Validation tab: 8.4% worst, hand derivation, honest discrepancy (Ch.9).
- 2:50 — Twin tab: recite the honest scope note (Ch.10).